MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE, ROCKET ASSISTED, XM548 LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-60932 (MU), dated 15 January 1968, and is approved for use by the US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Delete: "XM548" wherever referenced and substitute "XM548E1."

2. Applicable documents, drawings: Delete the following drawings:

"9209385 - Containers, Ammunition, Plastic, M-, for cartridge, 105MM, HE, RAP, XM548.

9209387 - Crate, Wood, Open, Wirebound for 105MM (Howitzer) Ammunition, in Plastic Container XM578."

PAGE 3

3.3.1:

Table I: Delete "*Pertains to both Composition B and Filler E."

Add new Table II as follows:

"TABLE II

	Inert Pad*
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings (sq in)	1/32
Projected length of any cavity excluding pipes, cracks and annular rings (in.)	1/4

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Piping cavities max projected area (sq in)	0
Piping cavities max projected width (in)	0
Cracks, max, projected width (in)	1/32
Annular rings, max. projected width (in)	0

*If any size cavity in inert pad is within 1/8" of the explosive surface, the cavity will be considered part of Segment A."

3.7: Delete in its entirety.

3.8.1.4.1: Delete in its entirety and substitute the following:

"3.8.1.4.1 The mean observed horizontal range of the test rounds shall be within 80 meters of the mean observed horizontal range of the reference rounds and the range standard deviation of the test rounds shall not be greater than 80 meters."

PAGE 7

4.2.1: Delete c, d, and e in their entirety and substitute the following:

"c. Composition B from the same grade from one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.

d. Supplementary charges from one manufacturer produced in accordance with the same drawing, same drawing revision, same operation, and same specification revision.

e. Motor body propellant grains of one lot interfix number from one manufacturer."

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4.2.2.4: 100 Percent Inspection

Major Defect 101: Add "Max" after word "cavity."

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Major Defect 103: Delete in its entirety.

Major Defect 104: Delete in its entirety.

Add new Minor Defect 204 as follows:

"100 Percent Inspection
Minor 204. Liner not coated for a full 360° around outside with silicone lubricant... Visual 04008."

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4.2.2.14:

Add new Major Defect 105 for 100 percent inspection as follows:

"105. Cement missing, motor body - warhead assembly threads... Visual 14010.

Delete major Defect 102 in its entirety.

Add new Major Defects 103 and 104 as follows:

"103. Concentricity of rotating band with warhead bourrelet...Gage 14007

104. Diameter of bourrelet (max)...Gage 14009."

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4.2.2.19:

Add new Major Defect 102 as follows:

"102. Shipping plug gasket missing... Visual 19015."

Delete Minor Defects 201 and 204 in their entirety.

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4.2.2.20: Delete in its entirety.

4.2.2.21: Delete in its entirety.

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4.2.2.22: Delete in its entirety.

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4.2.3.2: Delete in its entirety and substitute the following:

"4.2.3.2 Motor body. Warhead disassembly torque (see 3.5) - Major defect - Code No. 14011. Each projectile shall be inspected 100 percent for this defect. Any projectile which fails to comply with the applicable disassembly torque requirement shall be classed defective and removed from the lot or reworked to meet the requirements. Reworked projectiles shall be assembled as per drawing requirements and retested to determine compliance with the disassembly torque requirements."

PAGE 20

4.2.3.5: Delete in its entirety.

4.2.3.5.1: Delete in its entirety.

4.2.3.5.2: Delete in its entirety.

PAGE 21

4.2.3.6.1.1: Delete in its entirety and substitute the following:

"4.2.3.6.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, 30 sample cartridges shall be selected at random from each lot for this test, and fired with the rocket selector cap off (rocket motor to function). If a misfire or a critical defect occurs, the lot shall be rejected. In addition, if the standard deviation horizontal range multiplied by .77 or the muzzle velocity standard deviation multiplied by .77 exceeds the requirements specified in 3.8.1.4.1 or 3.8.1.4.3 or the mean range or muzzle velocity falls outside the limits specified in 3.8.1.4.1 or 3.8.1.4.3, the lot shall be rejected. The test shall be performed as specified in 4.3.3."

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4.2.3.6.1.2: Delete in its entirety and substitute the following:

"4.2.3.6.1.2 After three consecutive lots have complied with the criteria of 4.2.3.6.1.1, 20 sample cartridges shall be selected at random from each lot for this test, and fired with the rocket selector cap off (rocket motor to function). If a misfire or a critical defect occurs, the lot shall be rejected. In addition, if the standard deviation in the horizontal lot range multiplied by .73 or the muzzle velocity standard deviation multiplied by .73 exceeds the requirements specified in 3.8.1.4.1 or 3.8.1.4.3 or the mean range or muzzle velocity falls outside the limits specified in 3.8.1.4.1 or 3.8.1.4.3, the lot shall be rejected. The test shall be performed as specified in 4.3.3. If three successive lots fail to meet the requirements, the provisions of 4.2.3.6.1.1 shall be reinstated."

4.2.3.6.2.1: Delete in its entirety and substitute the following:

"4.2.3.6.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, 20 sample cartridges shall be selected from each lot for this test. If any cartridges fail to comply with any of the safety requirements specified, the lot shall be rejected. The test shall be conducted as specified in 4.3.3."

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4.2.3.6.2.2: Delete in its entirety and substitute the following:

"4.2.3.6.2.2 After three consecutive lots have complied with the criteria of 4.2.3.6.2.1, a sample of 10 cartridges shall be selected from each lot for this test. If any cartridge fails to comply with any of the safety requirements specified the lot shall be rejected. If three successive lots fail to meet these requirements, the provisions of 4.2.3.6.2.1 shall be reinstated. The test shall be conducted as specified in 4.3.3."

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Add new paragraph 4.2.3.7 as follows:

"4.2.3.7 Diameter of Cartridge case mouth for fit to projectile (see dwg. 9212376) - Major defect - Code No. 17004. The cartridge cases shall be gaged 100 percent for profile and diameter of mouth of case. Any cartridge case which fails to comply with the applicable requirements shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.2.4 (NOTE: This inspection shall not be required if it is determined that the cartridge cases have been previously gaged 100 percent for these characteristics.)."

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4.3.2: Delete in its entirety.

4.3.3.1: Delete in its entirety and substitute the following:

"4.3.3.1 Functioning - The cartridges for this test including 15 reference rounds for tightened inspection (see 4.2.3.6.1.1) or 10 reference rounds for normal inspection (see 4.2.3.6.1.2) shall be conditioned at 70 plus or minus 2 degrees Fahrenheit and fired within three minutes after removal from the conditioning box. The test and reference cartridges shall be fired in an equally distributed alternate manner in a howitzer equipped with an M103 or an M137 cannon using a quadrant elevation to assure near maximum range (i.e., 51 degrees) with a zone seven (7) XM176 propelling charge. The individual velocities and impact ranges shall be obtained and recorded. The standard deviation and the arithmetic average of the impact range shall be calculated and recorded. In addition, the individual pressure, the elapsed time to rocket functioning, elapsed time to rocket burnout and velocity versus time in 0.2 second intervals from 2 seconds before rocket ignition to 2 seconds after rocket burnout shall be obtained and recorded. Radar shall also be used to observe for metal parts security and in case of short rounds or unobserved dud rounds, radar magnetic tape records of such rounds along with the three preceeding and three following rounds shall be forwarded for analysis to Commander, USA ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-Q. For information purposes, a comparison of the mean corrected range of the test rounds with the range specified in the firing table shall be

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computed. The mean corrected range shall be based upon correction factors as presented in the firing table with the meteorological correction being derived from a computer analysis of the trajectory utilizing meteorological data taken in the midst of the test to a height of 15,000 feet."

4.3.3.2: Delete in its entirety and substitute the following:

"4.3.3.2 Safety. The cartridges shall be divided into two equal groups. Group I shall be conditioned at -65°F and fired at charge 7 in a M103 or M137 cannon. Group 2 shall be conditioned at 145°F and fired with the propellant adjusted to give 37,000 psi chamber pressure in a M103 or M137 cannon. Observations shall be made for premature burst, indication of metal parts separation in the gun bore or in flight, rocket motor functioning and range. Radar shall be used to observe for metal parts security, and in case of rounds exhibiting an unusual flight, radar magnetic tape records of the rounds in question along with the three preceeding and the three following rounds shall be forwarded for analysis to Commander, USA ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-Q."

PAGE 26

5.1: Delete in its entirety and substitute the following:

"5.1 Packaging - Level A. The containers shall be packed in accordance with Dwg. 9213636.

5.2: Delete in its entirety and substitute the following:

"5.2 Packing - Level A. The containers shall be packed in accordance with Dwg. 9213637.

5.3: Delete in its entirety and substitute the following:

"5.3: Marking. The containers shall be marked in accordance with Dwg. 9213636 and the wood boxes shall be marked in accordance with Dwg. 9213637."

Custodian:
Army-AR

Preparing Activity:
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